

Tyler Takimoto

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EDUCATION

Bachelor of Science, Computer Engineering
University of Central Florida

May 2026
Orlando, Florida

WORK EXPERIENCE

Engineering Associate
Mitsubishi Power Americas

May 2025 – May 2026
Orlando, Florida

- Developed a computer vision model utilizing PyTorch and Python to automate repair image processing, achieving 99% model accuracy on all unseen images.
- Architected a PyQt database editor with Python, SQL, and ODBC to resolve cross-team workflow bottlenecks.
- Programmed a 3D visualization tool in Python using Open3D, Pandas, and NumPy to interpret spatial coordinate data, improving in-house report generation.

PROJECT EXPERIENCE

NIMBL, Senior Design

August 2025 – May 2026

- Designed and programmed a low-power pulse-oximeter using a TI MSP430 MCU in C++, selecting hardware that successfully balanced processing capabilities with strict power and budget constraints.
- Developed custom hardware abstraction libraries to interface a MAX32664 biometric sensor, a CC1101 radio module, and an HT1261 LCD driver, enabling real-time heart rate and SpO2 tracking.
- Collaborated within a cross-function team to author a 120-page technical design report, detailing our thought processes, hardware choices, and team goals to document our project.

fAlnman, Knight Hacks VIII

October 2025

- Directed a development team through a 36-hour hackathon, designating roles based on technical expertise and managing important milestones to ensure effective project progress.
- Deployed the AI learning platform to a DigitalOcean server, creating a custom domain with HTTPS and Unicorn to provide a globally accessible application.
- Engineered a conversational AI listener with Google Gemini API and a database to store user sessions, driving context-aware study interactions based on the Feynman Technique.

Pocket Professors, Processes in Object Oriented Software

May 2025 – July 2026

- Directed the frontend development of a cross-platform card-collecting application, utilizing React TS for the web application and Flutter for the mobile app.
- Designed and implemented an intuitive user interface and integrated backend APIs, enabling users to create accounts and dynamically track their catalogue.
- Collaborated with a newly formed development team, meeting consistently to ensure smooth development and efficient project delivery.

Polly the Parrot, Knight Hacks VII

October 2024

- Created a talking AI parrot companion for the Knight Hacks VII 36-hour hackathon, collaborating on hardware design to achieve 2nd place for Best Hardware Hack.
- Integrated OpenAI ChatGPT and IBM Watson in Python to build a seamless Speech-to-Text and Text-to-Speech interaction loop, creating an interactive pirate-parrot persona focused on child engagement.
- Programmed hardware-software communication with pyFirmata to synchronize audio with the beak.

SKILLS

- **Back-End:** Python, C, C++, Java, Bash, SQL
- **Front-End:** HTML, CSS, JavaScript, TypeScript, React, Flutter, Tailwind CSS
- **Software:** Linux, Docker, GitHub, YAML, Machine Learning, Fusion360, Microsoft 365